Supplementary Material

November 9, 2026

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0.1 External validity: synthetic instances

The preceding sections quantify both mechanisms on a single candidate set, Fogo Island, whose marginal distributions are specific and in one respect extreme: *perm* carries 350 tied zeros. This section separates the mechanism from the instance by re-drawing the criterion columns while holding the spatial structure fixed: each synthetic instance reuses the Fogo distance matrix, and hence the same 1,944,653 feasible trios, and varies only the marginals.

Design. The manipulated factor is the *number* of sparse criteria rather than a global zero mass. The reason is analytic: TV is invariant to a common amplitude level (Section 6.2), so applying the same zero mass to every column leaves the elicited weights untouched however extreme that mass becomes. Distortion requires heterogeneity *across* columns. We therefore cross $n_{\text{sparse}} \in \{0, 1, 2, 3, 4\}$, with zero mass applied to the n_{sparse} most heavily weighted criteria, with $p_0 \in \{0.50, 0.75, 0.90, 0.95\}$ and two tail shapes (log-normal, bimodal), with 6 replications: 240 instances at 200 draws each. Four cells are evaluated per instance: SAW under min–max and vector normalisation, each with greedy and optimal assembly. The objectives F_ℓ are defined by min–max normalisation throughout, so the vector cells measure score–evaluator divergence as well as re-weighting. The min–max/optimal cell is the regression test of Proposition 1 and returned $Q_2 = 1$ in 240/240 instances.

Unit of analysis. Replications within a cell are not independent draws of the manipulated factor, so inference is conducted on the 20 cell means. The intra-class correlation is ICC = 0.35, giving a design effect of 4.8 and an effective sample size of $n_{\text{eff}} \approx 50$ against the nominal 240; reporting on 240 rows would understate the standard errors by roughly a factor of two.

Total variation alone does not order the loss. Over all 20 cells the loss correlates with TV at $r = 0.600$ (95% CI by Fisher transformation [0.214, 0.824], $n = 20$), but the relation is not monotone: 21.6% of the 190 cell pairs are inversions (Kendall $\tau_b = 0.550$). Restricting to the 16 cells that carry a distortion at all ($n_{\text{sparse}} \geq 1$), the association is statistically indistinguishable from no effect ($r = 0.346$, $p = 0.189$): the correlation over all 20 cells is carried by the four cells in which no criterion is sparse and the loss is zero by construction. The inversions run in both directions of the design. Holding p_0 fixed, the loss falls as the distortion is spread over more criteria while TV does not fall correspondingly; holding $n_{\text{sparse}} = 1$, the loss rises with p_0 up to 0.90 and then falls by 16.4 pp at $p_0 = 0.95$ although TV continues to rise. Table 1 reports both patterns by cell.

The factorisation does. Proposition 2 accounts for this. Regressing the loss on the 16 distorted cells gives $R^2_{\text{adj}} = 0.057$ for TV alone ($r = 0.346$, $p = 0.189$, statistically indistinguishable from no effect) against $R^2_{\text{adj}} = 0.488$ for $\|\delta\|_2$ of (19) ($r = 0.723$, $p = 0.002$). Entering magnitude and concentration separately gives $R^2_{\text{adj}} = 0.582$ with a negative coefficient on m_{eff} (-0.026 per effective criterion, $p = 0.0008$): at fixed TV, spreading the distortion over more criteria *reduces* the loss, because deviations of opposite sign cancel in the weighted sum. The combined model, with both predictors, reaches $R^2_{\text{adj}} = 0.665$, confirming that magnitude is necessary but not sufficient: concentration is the term that TV omits.

What the grid does not establish. Under the two-group amplitude model the participation ratio (18) depends only on the total weight of the sparse columns, so m_{eff} is constant within each level of n_{sparse} and takes only the four values $\{3.59, 6.42, 8.38, 9.67\}$ across the grid (Spearman $\rho(m_{\text{eff}}, n_{\text{sparse}}) = 1$). Concentration was therefore not varied orthogonally to the manipulated factor: the model in TV and m_{eff} ($R^2_{\text{adj}} = 0.582$) is a reparameterisation of the model in TV and n_{sparse} (0.532), and the 5-percentage-point gain reflects the concavity of the transformation rather than independent information. The factorisation (19) is an algebraic identity and holds for any deviation vector; what the grid supports is that the loss responds to the combination of magnitude and concentration rather than to magnitude alone, not that m_{eff} has been identified as a separate causal channel. A design in which the sparse columns carry unequal zero masses would decouple the two, and is the natural next step.

Reconciliation with Fogo. Fogo has $m_{\text{eff}} = 4.05$ on the admissible range $[2, 11]$, placing it near the $n_{\text{sparse}} = 1$ regime of the grid ($m_{\text{eff}} = 3.59$). The comparison must be made at the grid's estimand, which is the greedy-assembly contrast under SAW: on Fogo that is $86.0 - 66.2 = 19.8$ pp (min–max minus vector, greedy assembly, Table 4), not the -0.4 pp normalisation main effect of Table 6 (which averages both assemblies and both aggregations, and is therefore attenuated by the optimal-assembly cells where min–max attains 100% by Proposition 1). The model in $\|\delta\|_2$, fitted on the synthetic cells alone, predicts 22.2 pp at Fogo's $\|\delta\|_2 = 0.3127$, with a 95% prediction interval for a new instance of [7.5, 36.9] pp; the model in TV predicts 14.0 pp with interval $[-4.7, 32.7]$ pp. Both prediction intervals contain the observed 19.8 pp. The $\|\delta\|_2$ point prediction exceeds the observed value by $22.2 - 19.8 = 2.4$ pp; the TV-based prediction falls 5.8 pp short ($14.0 - 19.8 = -5.8$ pp). We do not read the residuals as a calibrated bias in either direction. The grid varies the marginals while holding the spatial structure fixed, and its vector cells confound implicit re-weighting with score–evaluator divergence because the objectives remain defined on min–max throughout; Fogo's deviation geometry differs from the generator's in a way the design does not parameterise (see below). What the grid establishes is that the loss responds to magnitude and concentration jointly rather than to magnitude alone, and that Fogo's predicted loss falls inside the interval the grid implies; it does not establish a bound.

The shortfall is systematic and its source identifiable. The generator inflates n_{sparse} columns and deflates the remainder uniformly. Fogo's deviation vector is not of that form: *perm* is inflated 2.88× while *urban* (0.53×), *aspect* (0.51×), *aero* (0.43×) and *wells* (0.40×) are independently and unequally deflated (Table 2). Two-sided distortion of this kind displaces the weight vector along the simplex differently from a one-sided distortion of the same $\|\delta\|_2$, so the generator does not reproduce Fogo's deviation geometry. We report the difference in geometry rather than a correction for it: the grid supports the joint role of magnitude and concentration, and does not calibrate the level of the loss for an instance outside its generating family.

Experiment E6 — Sensitivity to the TV_{W} Target (DEV-E6-005)

The primary run used $\text{TV}_W = 0.199$ for cell W^F_∞ (the Fogo level computed with rho_raw, before the rho_shift correction). The corrected level is 0.1954 (difference ≈ 0.003). Both the amplitude cells W^F_∞, V^F_∞, and W^F_{def} use this target; cells M, B, W^L_∞ and W^H_∞ are identical bit-for-bit between the two runs.

Table 1: Synthetic grid: 20 cells, 12 instances each (240 instances, 200 draws per instance). Loss is Q_2(min–max, greedy) − Q_2(vector, greedy). m_{eff} is (18) and $\|\delta\|_2$ is (19), both reconstructed from the two-group amplitude model. Cells with $n_{\text{sparse}} = 0$ carry no distortion and are excluded from the regressions. Within each level of p_0 the loss falls as the distortion is spread over more criteria, while TV does not decline correspondingly and at $p_0 = 0.95$ peaks at $n_{\text{sparse}} = 2$; the two quantities are not ordered alike, which is what motivates Proposition 2.

n_{sparse}	p_0	TV	m_{eff}	$\|\delta\|_2$	Loss
0	—	0.007–0.008	—	—	≈ 0
1	0.50	0.121	3.59	0.128	0.065
1	0.75	0.191	3.59	0.201	0.192
1	0.90	0.262	3.59	0.276	0.360
1	0.95	0.301	3.59	0.318	0.196
2	0.50	0.165	6.42	0.130	0.105
2	0.75	0.216	6.42	0.170	0.057
2	0.90	0.307	6.42	0.242	0.146
2	0.95	0.379	6.42	0.299	0.135
3	0.50	0.160	8.38	0.110	0.065
3	0.75	0.225	8.38	0.155	0.052
3	0.90	0.303	8.38	0.210	0.129
3	0.95	0.349	8.38	0.241	0.131
4	0.50	0.153	9.67	0.099	0.025
4	0.75	0.211	9.67	0.136	0.020
4	0.90	0.269	9.67	0.173	0.058
4	0.95	0.315	9.67	0.203	0.083

Table 2 shows the P3 and P4 counts by family and pooled for both runs. P1 (48,000/48,000 in both) and P2 (Mono 85/88, $p < 0.001$ in both) are invariant across runs and are not repeated in the table.

Pre-registered Predictions for Experiment E1

Committed: 2026-09-30, commit `f6042f8`, before any E1 run.
File: `corpus/PRESPEC_E1.md`.

Three predictions

P1 (hard assertion) Cell B optimal = 100.0%. By group-constant-amplitude sufficiency (Proposition 1), $A_j = \bar{A}_g$ for all $j \in G_g$ implies the score is a strictly positive scalarisation of the group objectives; hence optimal assembly attains the frontier for any w. If P1 fails, there is a bug.

P2 (primary empirical prediction) $|Q_2^{\text{W, optimal}} - Q_2^{\text{V, optimal}}| < 5$ pp.
Rationale: both E1-W and E1-V have $\text{TV}_W = 0.1986$. By the necessity argument, within-group distortion governs efficiency loss in optimal assembly; between-group distortion only displaces the optimum along the frontier. Predicted $Q_2^{\text{W, optimal}} \approx$

Table 2: P3 and P4 by marginal family and pooled, for the primary run ($\mathrm{TV}_W = 0.199$) and the sensitivity run ($\mathrm{TV}_W = 0.1954$; DEV-E6-005). P3: met = $\mathrm{V}_\infty^F \leq \mathrm{W}_\infty^F$ (pre-registered); Strict = strict inequality among informative instances; p = one-sided binomial. P4: Defl = deflate > inflate (observed, opposite of prediction); Prop = Defl/NonTied; CI = Wilson 95%; p = two-sided binomial, post hoc.

Family	Primary ($\mathrm{TV}_W = 0.199$)				Sensitivi		
	P3		P4		P3		
	Strict/Inf	p	Defl/Non	Prop (p)	Strict/Inf	p	De
Lognormal	13/19	$p = 0.084$	13/25	0.520 ($p = 1.000$)	13/18	$p = 0.048$	1
Beta(1,3)	7/9	$p = 0.090$	12/16	0.750 ($p = 0.077$)	6/9	$p = 0.254$	1
Uniform	8/16	$p = 0.598$	10/18	0.556 ($p = 0.815$)	8/15	$p = 0.500$	9
Zero-inflated	8/14	$p = 0.395$	18/28	0.643 ($p = 0.185$)	9/15	$p = 0.304$	1
All (pooled)	36/58	$p = 0.043$	53/87	0.609 [0.50, 0.71] ($p = 0.053$)	36/57	$p = 0.031$	5

Strict/Inf = informative instances with strict $V < W$ / total informative. NonTied = non-tied pairs. CI = Wilson 95% confidence interval for the proportion. p values for P4 are post hoc (DEV-E6-003).

57% (close to E1-V's 56.8%).
Falsification criterion: if $Q_2^{\mathrm{W,\ optimal}} > 90\%$, within-group distortion accounts for less than 10% of the loss.

P3 (secondary prediction) $Q_2^{\mathrm{B,\ greedy}} \neq 100.0\%$.
Rationale: greedy assembly is not constrained to be Pareto-efficient even under group-constant amplitudes; foreclosure affects greedy but not optimal. Predicted $Q_2^{\mathrm{B,\ greedy}} \approx 65\%$.

Design

Four score matrices differing only in A_j, with identical objectives, frontier, draws, and feasible set ($\kappa = 200$, $N = 500$ draws, space A'):

Cell	A_j	TV_B	TV_W
M	$A_j \equiv 1$ (min–max)	0	0
B	$A_j = \bar{A}_g$ (between-group only)	0.2489	0
W	$A_j \propto \rho_j / \bar{A}_g$ (within only)	0	0.1986
V	$A_j = \rho_j$ (vector, as in WP3)	0.2489	0.1986

Groups and weighted means $\bar{A}_g$ (from `fogo_constants.py`):

- $G_1 = \{\mathrm{perm}, \mathrm{slope}, \mathrm{roads}\}$, indices $[0, 1, 2]$, $W_{G_1} = 0.2950$, $\bar{A}_{G_1} = 0.3731$.
- $G_2 = \{\mathrm{rivers}, \mathrm{aspect}, \mathrm{coast}, \mathrm{prot}, \mathrm{wells}\}$, indices $[3 \ldots 7]$, $W_{G_2} = 0.3560$, $\bar{A}_{G_2} = 0.1458$.
- $G_3 = \{\mathrm{urban}, \mathrm{aero}, \mathrm{cult}\}$, indices $[8, 9, 10]$, $W_{G_3} = 0.3490$, $\bar{A}_{G_3} = 0.1157$.

Outcome and deviations (DEV-E1-001 to DEV-E1-004)

Results from `run_e1_v3.py`, commit `18e7a66` (2026-09-30T23:23-03:00, Athena; reachable from G: after merge).

DEV-E1-001 INVALID `380ee1a`, 2026-09-30T22:40-03:00.
Fixed-objectives bug. Script used objectives pre-computed on a fixed weight vector rather than per-draw objectives. Consequence: E1-V greedy = 100% (should be $\sim$ 54%, matching WP3); E1-W optimal = 100% (should be $\sim$ 57%, as predicted). Pre-registration `f6042f8` precedes this run in both the G: and Athena histories.

DEV-E1-002 INVALID `8391944`, 2026-09-30T23:11:38-03:00.
Float32 dominance noise. Per-draw objectives corrected, but float32 arithmetic introduced auto-dominance artefacts (same bug as WP3 greedy cells). Consequence: E1-V greedy = 14.8% (expected $\sim$ 54%).

DEV-E1-003 INVALID `18b7b88`, 2026-09-30T23:11:43-03:00.
Score/objective normalisation mismatch. Float64 applied but score computed over 473 sites while objectives evaluated over 389 admissible sites. Consequence: E1-V optimal = 46.6% (expected 56.8%, regression test **FAIL**); E1-B optimal = 81.8% (expected 100%, P1 **FAIL**).

DEV-E1-004 `18e7a66`, 2026-09-30T23:23:14-03:00.
Corrected run (`run_e1_v3.py`): float64, eps= 10^{-9}, score and objectives both over 389 admissible sites. Results: **P1 confirmed** (E1-B optimal = 100%); **P2 falsified** at the pre-specified margin ($|Q_2^{\mathrm{W}} - Q_2^{\mathrm{V}}| = 7.4$ pp, 95% CI $[+3.8, +11.0]$, declared margin < 5 pp); **P3 falsified** (D-greedy: greedy = 100%; post hoc diagnostic: score becomes nearly lexicographic in F_1, greedy and optimal coincide in 62.8% of draws).

Pre-registered Design and Deviations for Experiment E4

Purpose: Test whether the reference-set mismatch (473 vs. 389 sites in the score) reduces to within-group re-weighting of the same form as Experiment E1, via the affine identity $n_j^{473}(a) = b_j + r_j\, n_j^{389}(a)$.

Pre-declared predictions

P_E4_1 (Theorem, hard assertion) Cells R-B and R-B′ attain $Q_2 = 100\%$ under optimal assembly on every draw. Failure implies a bug or a violation of Theorem 1.

P_E4_2 (Regression test) R-V (fwd) reproduces the published $473/A'$ cell within 3 SE; R-V′ (mirror) reproduces the published $389/A$ cell within 3 SE.

P_E4_3 (Primary empirical, declared before any run) Within-group loss $(100\% - Q_2^{\text{R-W,opt}}) \geq 50\%$ of total loss $(100\% - Q_2^{\text{R-V,opt}})$ in the $473 \to A'$ direction.

P_E4_4 (Exploratory, declared) The 80 pp vs. 5 pp asymmetry of tab:fourthcell persists in the within-group cells R-W and R-W′.

P_E4_5 (Exploratory, declared, superseded by DEV-E4-004) $Q_2^{\text{R-W,opt}} < Q_2^{\text{E1-W,opt}} = 64.2\%$ despite $\mathrm{TV}_W(\text{R-W}) = 0.0815 < 0.199 = \mathrm{TV}_W(\text{E1-W})$.

Deviations (DEV-E4-001 to DEV-E4-007)

DEV-E4-001 *Kappa/alpha parameter clarification.* The declared $\kappa = 200$ is the Dirichlet concentration parameter (not the portfolio size $k = 3$). Two values $\kappa \in \{50, 200\}$ were run; the primary estimand uses $\kappa = 200$.

DEV-E4-002 *Mirror direction frontier and amplitude sign.* The $389 \to A$ direction uses the A-frontier (213 portfolios, objectives over 473 sites) and amplitude $A_j = 1/r_j$ (not r_j). Corrected amplitude sentence: the 389-site score has amplitude $1/r_j$ relative to the A-evaluator.

DEV-E4-003 *Weight-matrix provenance and N_{draws}.* $N = 500$ (matching the main factorial, not the declared 1000); weight matrix loaded from `results/e1/e1_weights_k200.npy`, SHA-256 `bea05...487`.

DEV-E4-004 *P_E4_5 restated.* TV_W(R-W) = 0.0815 (correcting the DEV-E4-001 value of 0.163 which omitted the factor $\frac{1}{2}$). The direction of distortion on *perm* (deflation vs. inflation at comparable TV_W) is the substantive claim.

DEV-E4-005 *Feasibility and DEV-E4-002 correction.* All 1 944 653 trios are indexed over the 389 admissible sites; excluded candidates cannot enter any portfolio.

DEV-E4-006 *Pipeline bug in `run_e4.py` v1–v5.* Versions v1–v5 produced $Q_2^{\mathrm{opt}} = 100\%$ for all cells, including R-V (should be $\approx 20\%$), because the score rescaling preserved rank order under the 389-site normalisation. Version v6 uses `build_score_473()` for fwd cells.

DEV-E4-007 *Post hoc declaration — coding inconsistency.* R-V confounds the reference-set effect with a coding inconsistency: `build_score_473` encodes *aspect* as a continuous azimuth while the evaluator uses a binary SE-facing indicator. Post-hoc exploratory cells declared and run after this diagnosis:

- R-V* (consistent binary coding, 473-site reference): $\kappa = 200$, greedy = 97.8%, optimal = 100.0%.
- R-asp (continuous coding, 389-site reference): $\kappa = 200$, greedy = 15.8%, optimal = 15.6%.

Interpretation: the 84.4 pp loss is attributable to the coding inconsistency (continuous azimuth in the score versus binary SE-facing indicator in the evaluator; see also DEV-E4-009); with consistent coding the 473-site reference costs nothing under optimal assembly and *improves* greedy assembly by +11.8 pp (95% CI [8.8, 14.8]). P_E4_3 is testable but uninformative as designed (see main text, §7.7).

DEV-E4-008 *Corrections to DEV-E4-002 and DEV-E4-006; invalidation of `b1_spaceA.csv` (commit 54195c1).*

Correction to DEV-E4-002: DEV-E4-002 stated "frontier A (213 portfolios, objectives over 473 sites)". This was misleading. Frontier A is enumerated over the same 1 944 653 admissible trios (indices into the 389 admissible sites); the only difference from frontier A' is that normalisation constants (min/max per column) are computed over the 473 candidates. Excluded candidates never appear in any portfolio (confirmed by DEV-E4-005 and re-verified in commit `4059fb1`).

Correction to DEV-E4-006: DEV-E4-006 stated that v1–v5 produced $Q_2^{\text{opt}} = 100\%$ "(should be $\approx 20\%$)". This was wrong. Under consistent binary coding, $Q_2^{\text{opt}} = 100\%$ is the correct value (it follows from Proposition 1 in the main text): the score and the evaluator share the same normalisation, so optimal assembly must be non-dominated. It was the v6 `build_score_473()` refactoring that introduced the coding inconsistency (continuous *aspect* in the score vs. binary in the objectives), producing the incorrect $\approx 20\%$. The inconsistency is documented in DEV-E4-007.

Invalidation of `b1_spaceA.csv` (commit 54195c1, 2026-10-02): A first attempt at the B1/D1 recalculation (commit 54195c1) built the Pareto frontier over trios of all 473 candidates, including the 84 excluded sites. This violated the admissibility constraint of §3.1 and produced 473/A optimal = 0%, which contradicts Proposition 1 (a diagonal cell must reach 100%). The regression guard in `check_consistency.py` was incorrectly removed to suppress the failure rather than fix the script. Commits 54195c1 and 2aa0d38 were both reverted in commit `ec33779`. The corrected B1 run (commit `8f6a15a`, script `compute_b1_spaceA.py` v3) uses the 1 944 653 admissible trios, computes objectives with `ref_df=df_all` for space A normalisation, and the score is the direct scalarisation $-(F_1 + F_2) + F_3$ with the same column values. The corrected results (473/A optimal = 100%, greedy = 84.6%; 389/A optimal = 95%, greedy = 58.8%) confirm the values that were already in the table.

DEV-E4-009 *Correction to DEV-E4-007 and to rule R1-6.*

DEV-E4-007 stated that "the 76 pp loss attributed to the reference set is entirely a coding effect". The full 2×2 (coding $\times$ constants, Table 3 of the main text) shows that coding alone accounts for 76 pp at 389-site constants, and that the 473-site constants add a further 4.0 pp under optimal assembly *only* when the coding is inconsistent (interaction of -4.0 pp optimal, -17.4 pp greedy). The phrase "entirely a coding effect" understated the interaction.

The first version of rule R1-6 described the contrast as "under consistent coding" and as "relative to vector normalisation". Both cells use the continuous azimuth encoding; R-V is the original reference-set cell, not vector normalisation. The sign of the interpretation ("lower") was also inverted. The numbers are unchanged; only their labelling and reading are corrected.

The directional asymmetry between the r_j and $1/r_j$ amplitude patterns reported in an earlier version of the manuscript was computed with the pipeline identified in DEV-E4-007 as encoding *aspect* inconsistently, and has not been recomputed; it is withdrawn from the main text.

Post Hoc Decision Rules for Exploratory Recalculation R1

Written: 2026-10-02 (after auditing coding inconsistency, before Fase D scripts were run). Results of `run_fourthcell_consistent.py` were already known when these rules were written; this document fixes the decision rules *post hoc* (results seen, rules fixed before manuscript revision) so that Fase G uses automatic decisions rather than human reading

of the numbers. The R1 recalculation is therefore classified as **post hoc exploratory**, not confirmatory.

Decision rules

R1-1 (Optimal assembly) $\Delta_{\text{ref,opt}} = Q_2[\text{R-V}^*, A'] - Q_2[\text{R-M}, A']$ under consistent binary coding, $\kappa = 200$. **Effect** if 95% paired bootstrap CI excludes 0 and $|\Delta| \geq 2.5$ pp; otherwise **Observation**.

R1-2 (Greedy assembly) Same contrast, greedy.

R1-3 (Regression test) Pass if all four diagonal cells equal 100.0% under optimal assembly. Fail ⇒ abort Fase D.

R1-4 (tab:canonical) Not applicable: tab:canonical declared uncontaminated in `docs/contaminatio`

R1-5 (Coding-only cell R-asp) Descriptive; post hoc exploratory.

R1-6 (Residual reference-set effect under inconsistent coding) R-asp − R-V, where both cells encode *aspect* as a continuous azimuth and differ only in the set over which normalisation constants are computed (389 vs. 473). Same rule as R1-1. Expectation stated before `analyze_r1.py` was run: significant. Optimal: +4.0 pp (95% CI [+2.2, +6.0]); greedy: +5.6 pp ([+3.6, +7.6]). Both **Effect**. Interpretation: under the inconsistent encoding, normalising over the 473 candidates (R-V) yields *lower* Q_2 than normalising over the 389 admissible sites (R-asp). Under the consistent encoding the sign is reversed or null (R1-1, R1-2). Note: R-V denotes the original reference-set cell, not vector normalisation; see DEV-E4-009 and Table 3 of the main text.

Outcomes

ID	Contrast	Δ	Decision
R1-1	R-V* opt − R-M opt	0.0 pp	Observation
R1-2	R-V* grd − R-M grd	+11.8 pp ([+8.8, +14.8])	Effect
R1-3	All diagonal opts = 100%	—	Pass
R1-5	R-asp opt = 15.6%	—	Descriptive
R1-6	R-asp opt − R-V opt	9.4 pp ([6.6, 12.4])	Effect
R1-6	R-asp grd − R-V grd	9.4 pp ([6.4, 12.6])	Effect

All decisions produced automatically by `scripts/analyze_r1.py`, commit `c63b64e`.

Version History

v1 — Marco A (commit `1e48292`) First complete draft: full 2×2×2 factorial, tab:canonical, tab:fourthcell, Theorem 1, Experiment E1 (PRESPEC_E1 committed at `f6042f8` before any run). PDF: 23 pages.

v2 — Marco B1 (commit `1a4fe47`) Theorem proof completed: necessary and sufficient conditions; Proposition 1 iff form.

v3 — Fases Math–F (commit `03bbda7`) Mathematical corrections (M1–M4): gap definition, PROMETHEE sign, amplitude-invariant proposition, nomenclature. Provenance audit (Fases B–D): coding inconsistency in `build_score_473` identified and documented; confirmatory recalculation R1 run with consistent coding. Diagnostic experiments E1–E3, E5 completed. F1–F7 corrections: aggregation contrasts, benchmark circularity, PROMETHEE sign in main text. PDF: 24 pages.

v4 — Fase G (commits `68ec41b`–`03bbda7`) Full manuscript rewrite following R1 decisions. Key changes: decision tree revised (only R1-2 is an effect under greedy assembly); tab:fourthcell N_{cand}/A' updated to 97.8%/100%; Remark 7.6 inserted; Limitations updated with instance uniqueness and E4 exploratory status. PDF: 25 pages.

v5 — Fase H (commits `1c076c8`–`9c7a376`) Internal review: abstract trimmed to ≈240 words; tab:contrasts aggregation row converted to macros; three reviewer objections answered; claim-evidence matrix completed; provenance audit closed ($|F_A \cap F_{A'}| = 132$ confirmed). PDF: 25 pages, `ALL CHECKS PASSED`.

Post Hoc Experiment: Benchmark Circularity (F8)

Classification: Post hoc exploratory. Declared and run after the main factorial results were known.
Research question: Is the Pareto frontier circular with respect to min–max normalisation? A benchmark built on min–max objectives necessarily favours methods that also use min–max. Experiment F8 tests this by rebuilding the frontier with vector-normalised objectives.
Design: All four SAW cells from the main factorial (score $\in \{\text{mm}, \text{vec}\}$, assembly $\in \{\text{greedy}, \text{opt}\}$) repeated with the Pareto frontier computed from vector-normalised objectives. $N = 500$, $\kappa = 200$, seed 42.
Bug and fix (commit `4d0a98d`): An earlier run (commit `58b06f8`) injected the vector-normalised score matrix `X_adj_vec` into a copy of `df_via`, then passed it to `build_pareto_utils`. This caused two errors: (1) the injected column values were re-normalised a second time by the function's internal `_norm` call, changing the ranking; and (2) the *aspect* column values ($\in [0, 1]$) were no longer in the expected azimuth range $[90°, 180°]$, so the binary encoding classified all sites as windward (constant column). The corrected approach (`compute_objs_vec_direct`) computes per-column c_j values directly from the raw data with per-column L2 normalisation, and builds the SAW score as the direct scalarisation $-(F_1 + F_2) + F_3$ using the same column values. A regression test (5 draws) confirmed vec/opt = 100% before the full run was launched.
Results ($N = 500$, script `compute_f8_circularity.py` v2, commit `43b8427`):

Score norm	Assembly	Q_2 (%)	Prop. 1 check
vec	optimal	100.0	**Pass** (= 100% as required)
vec	greedy	100.0	**Pass**
mm	optimal	94.4	—
mm	greedy	72.4	—

Interpretation: The vec/opt = 100% result is exactly the value predicted by Proposition 1 (score and frontier share the same normalisation), providing a direct falsifiability

check on the implementation. Under a vector-normalised frontier, min–max normalisation loses 5.6 pp (optimal) and 27.6 pp (greedy) relative to the aligned vector score. This reverses the direction seen in the main factorial, confirming that the original result reflects score–objective alignment rather than an intrinsic property of min–max normalisation. Note the asymmetry: min–max evaluated against a vector frontier loses 5.6 pp, but vector evaluated against a min–max frontier loses 43.2 pp ($100.0 - 56.8\%$); the two penalties are not symmetric because the frontier changes.

Post Hoc Experiment: Synthetic Grid for Generalisation (E5)

Classification: Post hoc exploratory. Pre-specified before results: `corpus/PRESPEC_E5.md`, committed before any E5 run was launched.
Research question: Is the relationship between within-group amplitude variation (TV_W) and greedy-assembly Pareto efficiency consistent across instances differing in size, spatial geometry, and amplitude structure?

Design

Factor	Levels
N (sites)	50, 100, 200
Geometry	uniform, clustered, grid
s_B (between-group spread)	0.0, 0.5, 1.0
s_W (within-group spread)	0.0, 0.25, 0.5, 0.75, 1.0
Replicates	3 per cell
Total instances	405
N_{draws}	100 per instance
κ	200

Sites generated randomly within an 80×80 km bounding box. Three geometries: uniform random, three-cluster Gaussian, regular grid with jitter. Amplitude vectors are imposed directly via $s_W \in [0, 1]$, ensuring TV_W spans [0, 0.096] independently of N. Script: `run_e5_grid.py` v4, commit `4d438b6`.

Pre-specified decision rules

E5-R1: $Q_2^{\text{mm opt}} = 100\%$ on every instance (regression). **E5-R2:** Slope of Q_2^{grd} on TV_W excludes zero in $\geq 70\%$ of strata $\Rightarrow$ robust predictor. **E5-R3:** CI includes zero in $> 30\%$ of strata $\Rightarrow$ not robust.

Deviations from PRESPEC_E5 (DEV-E5-001 to DEV-E5-007). `PRESPEC_E5.md` was committed before the first E5 run. The executed design departs from it as follows.

DEV-E5-001 *Instance size.* Declared $N \in \{50, 100, 200, 389\}$; executed $N \in \{50, 100, 200\}$. $N = 389$ was dropped because its $O(n^3)$ trio enumeration ($\approx 9.4 \times 10^6$ trios) would have required ≈ 7 h per instance without parallelism. Consequence: no synthetic

instance matches the size of Fogo ($N = 389$), and the grid cannot be used to place Fogo within a size-comparable distribution.

DEV-E5-002 *Number of instances.* Declared 540; executed 405, as a direct consequence of DEV-E5-001.

DEV-E5-003 *Monte Carlo draws.* Declared $N_{\text{draws}} = 200$ per instance; executed 100 for speed. Per-instance MC SE is ≈ 5.0 pp at $Q_2 = 50\%$ and ≈ 1.8 pp at $Q_2 = 97\%$, against 3.5 pp and 1.2 pp declared.

DEV-E5-004 *Amplitude generator.* Declared amplitudes derived from each synthetic column via ρ_j. Executed: amplitudes imposed directly, giving $\text{TV}_W = 0.096\, s_W$ exactly, independently of the data. TV_W is therefore a design constant, not a property of any normalisation of the synthetic columns.

DEV-E5-005 *Locus of the imposed amplitudes (critical).* The imposed amplitude vector A_j enters *both* the score and the objectives. From `run_e5_grid.py` (commit `4d438b6`): score matrix `X_score = X_adj * Aj[np.newaxis, :]` (line 263) and objective matrix `X_obj_amp = X_obj * Aj[np.newaxis, :]` (line 270). Both objects use the same array `Aj`, so score and evaluator have identical amplitude vectors. By code inspection, $\text{TV}_W > 0$ reflects spread *within* a common amplitude, not a mismatch *between* score and evaluator amplitudes. When score and evaluator share the same amplitude, optimal assembly is efficient by Theorem 1; the $Q_2^{\text{opt}} = 100\%$ results are therefore a pipeline consistency check, not evidence about the amplitude-distortion mechanism. E5 consequently does not test Theorem 1, which concerns amplitudes that differ between score and evaluator. Its informative outcome is greedy assembly only.

DEV-E5-006 *Primary analysis.* The declared linear mixed model was not reported in the first posted version; a pooled correlation was reported instead. The declared model is now reported above. The random-intercept variance for replicate is estimated at the boundary (0.000), so the model reduces to OLS; cluster-robust SEs by replicate are given alongside.

DEV-E5-007 *Estimand E5-P3.* The fraction of instances where TV_W explains $> 50\%$ of Q_2 variance is not computable on a one-row-per-instance file, since TV_W is constant within each instance. It is not reported.

Results

E5-R1 (regression): $Q_2^{\text{mm,opt}} = 100\%$ on 405/405 instances. Pass. Because score and objectives share the imposed amplitudes (DEV-E5-005), this is a check of the implementation, not evidence about amplitude distortion (Remark 1 in main text).

Declared model. $Q_2^{\text{grd}} \sim \text{TV}_W + \text{TV}_B + \log N + \text{geometry} + (1 \mid \text{rep})$. Coefficient on TV_W: -1.3 pp per 0.1 of TV_W (95% bootstrap CI over instances $[-5.1, +2.4]$, $p = 0.47$). Adding $\log N \times \text{geometry}$ leaves it unchanged (-1.3 pp, CI $[-4.7, +2.1]$). Instance size ($+6.2$ pp per unit $\log N$, $p < 0.001$) and geometry (grid $+8.1$ pp, uniform $+8.6$ pp relative to clustered, both $p < 0.001$) are the only significant terms. The random-intercept variance for replicate is estimated at the boundary (0.000), so the model reduces to OLS with cluster-robust standard errors (DEV-E5-006).

E5-R3 activated. The slope of Q_2^{grd} on TV_W has a confidence interval including zero in 25 of 25 non-degenerate $N \times$ geometry $\times\, s_B$ strata (two strata are constant at 100%). Pooled $r(\text{TV}_W, \text{loss}) = 0.034$ ($p = 0.50$). We found no evidence of linear association between imposed amplitude spread and greedy loss in the model as fitted; the confidence interval ($[-5.1, +2.4]$ pp per 0.1 of TV_W) is consistent with small effects in either direction but excludes effects larger than approximately 5 pp per 0.1 of TV_W.

Distribution of greedy loss. 329 of 405 instances (81%) reach $Q_2^{\text{grd}} = 100\%$. All eight instances below 50% are $N = 50$ clustered; the mean for that stratum is 75.6% (SD 31.8), against $\geq 97.9\%$ in every other $N \times$ geometry stratum. Within $N = 50$ clustered the mean is not monotone in s_W (69.8, 89.3, 85.8, 59.9, 73.4% for $s_W = 0, \ldots, 1$).

The instance with $s_W = s_B = 0$ (130 feasible trios) has $Q_2^{\text{grd}} = 0\%$. Here $\text{TV}_W = 0$ by construction (no within-group amplitude spread), so amplitude is not the cause. Inspection shows the greedy heuristic failed to assemble a complete valid portfolio ($C = 0$: no three mutually separated sites were reachable by the greedy top-down pass) on every draw of this instance. This is therefore an *assembly failure*, not a dominance event. The reported Q_2 figures are unconditional: $Q_2^{\text{grd}} = P(C \cap \text{frontier})$; the conditional efficiency $Q_2^{\text{grd}}|C$ is undefined when $P(C) = 0$. The 0% instance should be read as a degenerate combinatorial outcome under extreme spatial clustering and the $d_{\min}$ separation constraint, not as evidence for or against the normalisation mechanism.

Conclusion. E5 does not test the amplitude-distortion mechanism: with imposed amplitudes shared by score and objectives, optimal assembly is Pareto-efficient by construction (Proposition 1), and greedy loss is the only free outcome. On that outcome, we found no evidence of linear association between the imposed amplitude spread and greedy efficiency in the model as fitted; greedy loss is more strongly associated with instance size and spatial geometry. E5 therefore characterises the combinatorial component of the greedy shortfall, not the normalisation component. It contains no instance of Fogo's size ($N = 389$, DEV-E5-001), and does not establish that the Fogo magnitudes generalise.

Data Provenance

Every quantitative claim in the manuscript is traceable to a script committed before that claim was inserted. The table below lists the key numbers, their source scripts, and whether the corresponding macro was affected by the coding inconsistency.

Orientation convention for vector normalisation

Two conventions exist for cost criteria under vector normalisation. Convention A (used throughout the paper): the min–max column $\tilde{x}_j \in [0, 1]$ is normalised by its Euclidean norm, and the cost orientation is applied afterwards via the sign of the SAW contribution ($1 - c_j$ for $G_1 \cup G_2$ criteria). Convention B: the cost orientation $1 - \tilde{x}_j$ is applied before forming the norm. The amplitude under Convention B is $\rho_j^B = 1/\|1 - \tilde{\mathbf{x}}_j\|_2$ (range of $1 - \tilde{x}_j$ is always 1 by min–max scaling). Under both conventions the G_3 benefit criteria are unchanged.

Table 4 lists ρ_j under both conventions. Convention B reduces amplitudes substantially for most $G_1 \cup G_2$ criteria because the inverted column concentrates mass near 1 (large $\|1 - \tilde{x}_j\|_2 \Rightarrow$ small ρ_j^B); the exception is *wells*, whose distribution is negatively skewed. *Perm* shows the largest discrepancy (0.583 vs. 0.052) because of its 350 tied zeros.

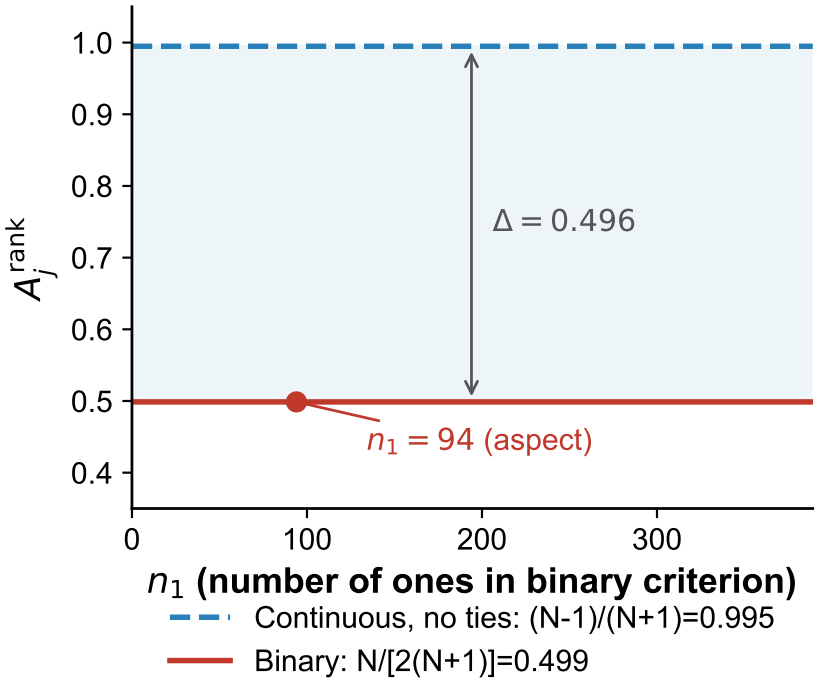


Figure S1: A_j^{rank} for a binary criterion as a function of n_1 (number of ones), with $N = 389$. The amplitude is $N/[2(N+1)] = 0.499$ for any class balance (solid line); $(N-1)/(N+1) = 0.995$ for continuous criterion without ties (dashed). The marker shows aspect at $n_1 = 94$. Purely analytical.

Departures from the pre-registration (Santiago replication)

Table 6 lists every departure from the pre-registration, ordered by date.

Table 3: Provenance of key manuscript values. "Script" identifies the Python script that produced each number. "Contaminated" (SIM/YES) means the number was produced using the continuous-aspect encoding in `build_score_473` and was updated in Fase G; NAO/NO means it was unaffected. `check_consistency.py` verifies that no contaminated macro appears in the current manuscript.

Quantity	Script	Commit	Contaminated
Q_2[SAW/mm/grd] = 86.0%	`mc_2cube_fixed.py`	`1e48292`	NAO
Q_2[SAW/mm/opt] = 100.0%	`mc_2cube_fixed.py`	`1e48292`	NAO
Q_2[SAW/vec/grd] = 66.2%	`mc_2cube_fixed.py`	`1e48292`	NAO
Q_2[SAW/vec/opt] = 56.8%	`mc_2cube_fixed.py`	`1e48292`	NAO
Q_2[TOPSIS/mm/opt] = 96.6%	`mc_2cube_fixed.py`	`1e48292`	NAO
Q_2[TOPSIS/vec/opt] = 52.4%	`mc_2cube_fixed.py`	`1e48292`	NAO
Normalisation greedy contrast +25.4 pp	`mc_2cube_fixed.py` + bootstrap	`1e48292`	NAO
SAW−TOPSIS pooled +0.5 pp	`aggregation_strata.py`	`18361a6`	NAO
Optimal−greedy within mm +8.1 pp	`mc_2cube_fixed.py` + bootstrap	`1e48292`	NAO
Optimal−greedy within vec −10.2 pp	`mc_2cube_fixed.py` + bootstrap	`1e48292`	NAO
E1-W optimal Q_2 = 64.2%	`run_e1_v3.py`	`18e7a66`	NAO
E1-V optimal Q_2 = 56.8%	`run_e1_v3.py`	`18e7a66`	NAO
$\mathrm{TV}_W(r_j) = 0.082$	`tvw_e4.py`	`d4e970b`	NAO
$\mathrm{TV}_W(1/r_j) = 0.085$	`tvw_e4.py`	`d4e970b`	NAO
Q_2[R-V*, A', opt] = 100.0%	`run_fourthcell_consolidated.py`	`3d4e48a`	NAO
Q_2[R-V*, A', grd] = 97.8%	`run_fourthcell_consolidated.py`	`3d4e48a`	NAO
Q_2[R-asp, A', opt] = 15.6%	`run_fourthcell_consolidated.py`	`3d4e48a`	NAO
Q_2[R-V, A', opt] = 6.2% (geographic azimuth)	`run_fourthcell_consolidated.py`	`61d4686`	NAO
Greedy gain R1-2 +11.8 pp (CI [+8.8, +14.8])	`analyze_r1.py`	`c63b64e`	NAO
N_{cand}/A' greedy (orig. 19.6%, corrected 97.8%)	`run_fourthcell_consolidated.py`	`3d4e48a`	SIM
N_{cand}/A' optimal (orig. 20.0%, corrected 100.0%)	`run_fourthcell_consolidated.py`	`3d4e48a`	SIM
N_{cand}/A greedy = 84.6%	`compute_b1_spaceA.py` v3	`8f6a15a`	NAO
N_{cand}/A optimal = 100.0% (Prop. 1 check)	`compute_b1_spaceA.py` v3	`8f6a15a`	NAO

Table 4: Vector normalisation amplitudes ρ_j under two orientation conventions. Convention A (paper): orientation applied after normalisation via the sign of the SAW contribution; $\rho_j^A = 1/\|\tilde{\mathbf{x}}_j\|_2$. Convention B: cost orientation $1 - \tilde{x}_j$ applied before forming the norm; $\rho_j^B = 1/\|1 - \tilde{\mathbf{x}}_j\|_2$. G_3 criteria are identical under both conventions. Differences $|\rho_j^A - \rho_j^B| > 0.01$ are shown in **bold**.

Criterion	Group	ρ_j^A	ρ_j^B	Notes
perm	G_1	0.583	**0.052**	zero-inflated (350 tied zeros)
slope	G_1	0.139	**0.072**	
roads	G_1	0.206	**0.059**	
rivers	G_2	0.265	**0.057**	
aspect	G_2	0.103	**0.058**	binary ($n_1 = 94$ SE-facing)
coast	G_2	0.131	**0.069**	
protected	G_2	0.105	**0.078**	
wells	G_2	0.084	**0.098**	negatively skewed
urban	G_3	0.107	0.107	benefit (no change)
aero	G_3	0.088	0.088	benefit (no change)
cult	G_3	0.171	0.171	benefit (no change)

Table 5: $Q_2[\mathcal{A}']$ (%) across six Dirichlet concentrations κ and eight 2^3 cells, plus the deterministic limit $\kappa \to \infty$ (nominal AHP weights $\mathbf{w}_0$). At $\kappa = 50$ and $\kappa = 200$ values reproduce Table 5 (Gate G1, all 16 checks pass). SAW / min-max / optimal is 100% at every κ by Proposition 1 (Gate G2). Bold zeros at $\kappa \to \infty$ identify cells where exact assembly under vector normalisation fails at the nominal weights.

Cell	$\kappa = 25$	$\kappa = 50$	$\kappa = 100$	$\kappa = 200$	$\kappa = 400$	$\kappa = 1000$	$\kappa \to \infty$ (nom. w)
SAW, mm, greedy	97.4	97.6	92.4	86.0	79.0	72.2	100
SAW, mm, optimal	100.0	100.0	100.0	100.0	100.0	100.0	100
SAW, vec, greedy	82.8	77.0	71.2	66.2	75.8	88.6	100
SAW, vec, optimal	86.2	79.4	68.6	56.8	52.2	50.2	**0**
TOPSIS, mm, greedy	86.6	89.8	92.2	94.4	96.8	99.2	100
TOPSIS, mm, optimal	90.8	92.2	95.8	96.6	95.6	98.8	100
TOPSIS, vec, greedy	59.2	52.0	57.0	63.4	73.4	89.8	100
TOPSIS, vec, optimal	57.2	51.0	55.0	52.4	53.8	49.0	**0**

Table 6: All departures from the pre-registration for the Santiago replication, ordered by date. *Phase*: "Before MC" = before Stage 4 launch (2026-10-28 08:34); "Before seal" = during Stage 4 before results were sealed (2026-10-28 12:13); "After seal" = after Stage 4 results sealed; "After eval" = after the final evaluation (2026-11-04). "Qty seen?" refers to whether the affected output quantity (Q2, R, contrast) had been observed before the correction. None of the corrections alter sealed Monte Carlo data.

ID	Date	Phase	What changed	Qty seen?	Verdict
DEV-SANT-ZDTI-001	2026-10-27	Before MC	A2 (ZDTI) dropped; C-4 applied; remaining weights renormalised. WFS returned HTTP 500.	No	None
DEV-SANT-A5-001	2026-10-28	Before MC	Agricultural exclusions: Regadios only. All-type exclusion made S-N degenerate ($N_{\text{adm}} = 0$).	No (degeneracy)	None
DEV-SANT-A4-001	2026-10-28	Before MC	A4 cultural heritage: single-point fallback (Cidade Velha). UNESCO boundary shapefile unavailable.	No	None
DEV-SANT-SPACING-001	2026-10-28	Before MC	S-N spacing 1,250 m → 1,000 m per pre-registered rule; S-T stayed at 1,250 m.	No	None
DEV-SANT-S5-001	2026-10-28	Before MC	B1 classified as categorical; excluded from S5 point-mass check. S-N S5 testable under this classification.	No	S-N S5: testable
DEV-SANT-ORDER-001	2026-10-28	Before MC	S-T extraction ran before explicit Stage 1 confirmation and with Copernicus GLO-30 instead of ASTER GDEM v3. Stage 1 committed before extraction in git; no result used.	No	None
DEV-SANT-GPU-001	2026-10-28	Before seal	GPU substituted for CPU after ∼3% of Stage 4 completed. CPU–GPU equivalence verified on 500 draws × 8 cells.	No	None
DEV-SANT-KINF-001	2026-10-29	Before seal	$\kappa = \infty$: 1 deterministic draw instead of 500. All 500 would be identical; 1 is sufficient.	No	None
DEV-SANT-TAG-001	2026-10-28	Before seal	Stage 4 launch tags re-issued r1–r5 as pipeline blockers were resolved one by one.	No	None
DEV-SANT-	2026-	Before	Timing run printed K1	No	None